BB-C-104B
March 1, 1971
SUPERSEDING
Fed. Spec. BB-C-104A
September 12, 1963

FEDERAL SPECIFICATION

CARBON DIOXIDE, TECHNICAL: SOLID

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type and one grade of solidified carbon dioxide commonly known as "dry ice".

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 6830

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:42Z
Check the source to verify that this is the current version before use.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The solid carbon dioxide, dry ice, shall be prepared from carbon dioxide "snow" by compressing into blocks or when specified (see 6.2) by extruding into pellets and shall be as specified herein.

3.2 Purity. The solid carbon dioxide shall be 99.95 percent by weight pure.

3.2.1 Residue. The residue after evaporation of the carbon dioxide shall not be greater than 0.05 percent by weight.

3.2.2 Color and odor. The solid carbon dioxide shall be snow white in color and free of odors.

3.3 Dimensions and weight. Unless otherwise specified (see 6.2), the solid carbon dioxide shall be in the form of 10-inch cubes, plus or minus 1/2-inch on a side, weighing at least 50 pounds each.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:42Z
Check the source to verify that this is the current version before use.

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105, inspection level S-2. For purposes of inspection all solid carbon dioxide offered for delivery at one time shall be considered a lot. Each block or a 50-pound quantity of the solid carbon dioxide shall be considered a unit of product.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

4.3.3 Test. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The solid carbon dioxide shall be examined for the following defects:

101. Evidence of discoloration.
102. Evidence of odor.
103. Dimensions and weight not as specified.

4.4.2 Test. A portion weighing not less than 5 kilograms shall be taken from each sample. The portion shall be weighed and placed in a container which has been dried to a constant weight at a temperature of 105° C. The portion shall be permitted to evaporate at room temperature. The container shall again be dried to a constant weight at 105° C. The weight of the residue shall be determined and the weight percentage of residue computed. All weight measurements required by the test shall at the least be accurate to the nearest 0.1 gram. Residue in excess of 0.05 percent by weight shall constitute failure of this test.

4.5 Inspection of preparation for delivery. The packaging and packing shall be examined to determine compliance with section 5 of this specification. The marking shall be examined to determine if any marking is missing, illegible, incorrect, or incomplete. Any one defect shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing (see 6.4). The solid carbon dioxide shall be packaged and packed in accordance with the supplier's commercial practice.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:42Z
Check the source to verify that this is the current version before use.

5.2 Marking. Packages shall be marked with the name of the material and the name of the manufacturer or supplier. In addition, the following caution notice shall appear on each package:

"CARBON DIOXIDE, SOLID (DRY ICE)
WARNING: EXTREMELY COLD (-109° F.)

Do not put in stoppered or closed containers.
Do not handle with bare hands; may cause severe burns.
Liberates heavy CO_2 gas which may cause suffocation.
Do not enter confined areas where used or stored until adequately ventilated."

5.2.1 Civil agencies. Shipment shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. Shipment shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Solid carbon dioxide covered by this specification is intended for use as a cooling agent and not for carbonation of beverages.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement documents:

(a) Title, number, and date of this specification.
(b) When pellets are required rather than blocks as specified in 3.1
(c) Size and weight of blocks required if other than as specified in 3.3.

6.3 Solid carbon dioxide ("dry ice") should be purchased on a net weight basis at the point of delivery.

6.4 The nature of the item is such that levels of packaging and packing are not considered appropriate. For purposes of packaging/packing level marking, the marking shall be A/A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:42Z
Check the source to verify that this is the current version before use.

MILITARY INTEREST:

Custodians:

Army - ME
Air Force - 68

User activities:

Navy - SH, YD, OS

Preparing activity:

Army - ME

CIVIL AGENCY INTEREST:

GSA

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies and other documents referenced herein. Price 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:42Z
Check the source to verify that this is the current version before use.